The class ψ_1 on $\overline{\mathcal{M}}_{1,n}$

Proof of Zvonkine's Proposition 2.22 · Lecture of 19 October 2026

Reference. D. Zvonkine, *An introduction to moduli spaces of curves and their intersection theory*: Proposition 2.22 is on printed p. 690 (p. 24 of the PDF). Zvonkine only says that the proof follows from Proposition 2.21 (pp. 689–690) and Proposition 2.17 (p. 688). The notes below fill in this argument.

Setup and statement

Cohomology has rational coefficients. We regard $\overline{\mathcal{M}}_{1,n}$ as a smooth Deligne–Mumford stack, so a point with automorphism group G has degree $1/|G|$. The class $\psi_1 = c_1(L_1)$ is the first Chern class of the line bundle whose fiber at $(C; x_1, \dots, x_n)$ is $T^*_{x_1} C$.

Boundary divisors. Let $\delta_{(\mathrm{irr})} \subset \overline{\mathcal{M}}_{1,n}$ be the closure of the locus of curves with a nonseparating node. For $S \subseteq \{1, \dots, n\}$ with $|S| \geq 2$, let $D_S \subset \overline{\mathcal{M}}_{1,n}$ be the closure of the locus of curves $E \cup R$, where E is a smooth curve of genus one, $R \cong \mathbb{P}^1$, the two components meet in a single node, and exactly the markings x_i with $i \in S$ lie on R:



(Vertices are components labeled by their genus.) The case $S = \{1, \dots, n\}$ is allowed, since E is stable with one special point. Both $\delta_{(\mathrm{irr})}$ and D_S are irreducible, being the images of the gluing maps from $\overline{\mathcal{M}}_{0,n+2}$ and from $\overline{\mathcal{M}}_{1,n-|S|+1} \times \overline{\mathcal{M}}_{0,|S|+1}$. Zvonkine's divisor $\delta_{(1)}$ is the locus where a separating node splits off a genus-zero part containing x_1, so its irreducible components are the D_S with $1 \in S$:

$$[\delta_{(1)}] = \sum_{\substack{S \subseteq \{1,\dots,n\} \\ 1 \in S,\ |S| \geq 2}} [D_S]. \tag{1}$$

In particular, $[\delta_{(1)}] = 0$ on $\overline{\mathcal{M}}_{1,1}$.

Proposition 2.22 (Zvonkine). *For every $n \geq 1$,*

$$\psi_1 = \frac{1}{12}\,[\delta_{(\mathrm{irr})}] + [\delta_{(1)}] \qquad \text{in } H^2(\overline{\mathcal{M}}_{1,n}; \mathbb{Q}).$$

Strategy. We use induction on n. Proposition 2.21 gives the case $n = 1$. For the inductive step, we pull both sides back along the map that forgets the last marking. Proposition 2.17 controls the left side, and the Lemma in Step 2 controls the right side.

Proof

Step 1: the case $n = 1$. The coarse moduli space of $\overline{\mathcal{M}}_{1,1}$ is the j-line $\mathbb{P}^1$. Hence $H^2(\overline{\mathcal{M}}_{1,1}; \mathbb{Q}) \cong \mathbb{Q}$, and integration is an isomorphism, so it is enough to compare degrees. On the right, $[\delta_{(1)}] = 0$, and $\delta_{(\mathrm{irr})}$ is the single point $[C_0]$, where C_0 is the nodal rational curve with one marking. Normalize C_0 as $\mathbb{P}^1$ with the branches of the node at $0, \infty$ and the marking at 1. An automorphism of C_0 is a Möbius transformation fixing 1 and preserving $\{0, \infty\}$, hence $z \mapsto z$ or $z \mapsto z^{-1}$. Thus $|\mathrm{Aut}(C_0)| = 2$. Together with Proposition 2.21, this gives

$$\int_{\overline{\mathcal{M}}_{1,1}} \psi_1 = \frac{1}{24} = \frac{1}{12} \cdot \frac{1}{2} = \frac{1}{12} \int_{\overline{\mathcal{M}}_{1,1}} [\delta_{(\mathrm{irr})}],$$

which proves the case $n = 1$.

Step 2: pulling back along the forgetful map. Fix $n \geq 1$, and let $p\colon \overline{\mathcal{M}}_{1,n+1} \to \overline{\mathcal{M}}_{1,n}$ forget x_{n+1} and stabilize. Objects on $\overline{\mathcal{M}}_{1,n+1}$ carry a prime, namely ψ_1', $\delta'_{(\mathrm{irr})}$, $\delta'_{(1)}$, and D'_T for $T \subseteq \{1, \ldots, n+1\}$.

The left side. By Proposition 2.17 with $i = 1$, $\psi_1' = p^*\psi_1 + [\delta_{(1,n+1)}]$. Here $\delta_{(1,n+1)}$ is the divisor of curves with a rational component whose only special points are one node, x_1, and x_{n+1}. Since this component has only one node, it is a rational tail, and the rest of the curve has genus one and carries all other markings. Hence $\delta_{(1,n+1)} = D'_{\{1,n+1\}}$, and

$$\psi_1' = p^*\psi_1 + [D'_{\{1,n+1\}}]. \tag{2}$$

The right side.

Lemma.

(a) $p^*[\delta_{(\mathrm{irr})}] = [\delta'_{(\mathrm{irr})}]$.
(b) *For* $S \subseteq \{1, \ldots, n\}$ *with* $|S| \geq 2$, $p^*[D_S] = [D'_S] + [D'_{S\cup\{n+1\}}]$.

Proof. Recall that p is identified with the universal curve over $\overline{\mathcal{M}}_{1,n}$. Thus the fiber of p over $(C; x_1, \ldots, x_n)$ is C itself. A smooth unmarked point $y \in C$ corresponds to $(C; x_1, \ldots, x_n, y)$, and in that case nothing is contracted. In particular, p is flat, and it is smooth at such points.

Let $D \subset \overline{\mathcal{M}}_{1,n}$ be a reduced irreducible divisor. Since p is flat, $p^*[D] = [p^{-1}(D)]$, and every irreducible component of $p^{-1}(D)$ dominates D. Thus the components can be read off from the fiber C over a general point of D. At a general point of each component, x_{n+1} is a general point of C, hence smooth and unmarked. So p is smooth there. A smooth pullback of a reduced divisor is reduced, so each component occurs with multiplicity one.

(a) A general $C \in \delta_{(\mathrm{irr})}$ is irreducible with one nonseparating node. Hence $p^{-1}(\delta_{(\mathrm{irr})})$ is irreducible. Its general point is C with x_{n+1} at a general point, and this is a general point of $\delta'_{(\mathrm{irr})}$.

(b) A general $C \in D_S$ is $E \cup R$ as in the figure. If x_{n+1} is a general point of E, the result is a general point of D'_S. If x_{n+1} is a general point of R, the result is a general point of $D'_{S\cup\{n+1\}}$. □

Step 3: bookkeeping. Summing (b) over all S in (1), we get

$$p^*[\delta_{(1)}] = \sum_{\substack{S\subseteq\{1,\ldots,n\}\\ 1\in S,\ |S|\geq 2}} ([D'_S] + [D'_{S\cup\{n+1\}}]).$$

Compare this with the expansion (1) of $[\delta'_{(1)}]$. Each $T \subseteq \{1, \ldots, n+1\}$ with $1 \in T$ and $|T| \geq 2$ belongs to exactly one of the following cases:

- $n+1 \notin T$. Then D'_T occurs once, as D'_S with $S = T$.
- $n+1 \in T$ and $|T| \geq 3$. Then D'_T occurs once, as $D'_{S\cup\{n+1\}}$ with $S = T \smallsetminus \{n+1\}$.
- $T = \{1, n+1\}$. Then D'_T does not occur, because it would require $S = \{1\}$.

Therefore

$$[\delta'_{(1)}] = p^*[\delta_{(1)}] + [D'_{\{1,n+1\}}]. \tag{3}$$

The same correction term appears in (2) and (3). On $D'_{\{1,n+1\}}$, the map p contracts the rational tail containing x_1, so this component is not seen by the pullback of $[\delta_{(1)}]$.

Step 4: conclusion. Assume the proposition holds on $\overline{\mathcal{M}}_{1,n}$. Then

$$\begin{aligned}
\psi_1' &= p^*\psi_1 + [D'_{\{1,n+1\}}] && \text{by (2)}\\
&= \tfrac{1}{12}\, p^*[\delta_{(\mathrm{irr})}] + p^*[\delta_{(1)}] + [D'_{\{1,n+1\}}] && \text{by the induction hypothesis}\\
&= \tfrac{1}{12}\, [\delta'_{(\mathrm{irr})}] + [\delta'_{(1)}] && \text{by Lemma (a) and (3).}
\end{aligned}$$

This is the proposition on $\overline{\mathcal{M}}_{1,n+1}$, which completes the induction. □